UFCFK1-15-0 - Fundamentals of data science 25feb_1

Take home assessment 40% (PSA)

Documentation 4

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Section : A

GIT : https://github.com/999-ztj/FODS_assignments

Assignment 4

Question 1

Program Description:

This program generates a NumPy array and performs basic operations: finding the sum, mean, largest and smallest values.

Screenshot:

```
Array: [10 20 30 40 50]

Sum of all elements: 150
Mean value: 30.0
Largest value: 50
Smallest value: 10
```

Question 2

Program Description:

This program takes a list of at least 12 numbers from the user,sorts the list, and performs slicing to extract elements at index ranges 3-6, 6-9, and 4-10.

Screenshot:

```
Enter at least 12 numbers separated by spaces:
1 2 3 4 5 6 7 8 9 10 11 12

Original list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
Sorted list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

Slice [3:6] (index 3 to 5): [4, 5, 6]
Slice [6:9] (index 6 to 8): [7, 8, 9]
Slice [4:10] (index 4 to 9): [5, 6, 7, 8, 9, 10]

Process finished with exit code 0
```

Question 3

Program Description:

This program creates an array of 12 random integers using NumPy,sorts the array, and reshapes it into a 3x4 matrix.

Screenshot:

```
Original random array: [46  5 45 37 38 81 65 74 13 56 24 56]
Sorted array: [ 5 13 24 37 38 45 46 56 56 65 74 81]

Reshaped into 3x4 matrix:
[[ 5 13 24 37]
 [38 45 46 56]
 [56 65 74 81]]

Reshaped into 4x3 matrix:
[[ 5 13 24]
 [37 38 45]
 [46 56 56]
 [65 74 81]]
```

Question 4

Program Description:

Matrix operations using NumPy with dimension validation.Performs addition, subtraction, and multiplication. Raises exceptions if dimensions are incompatible.

Screenshot:

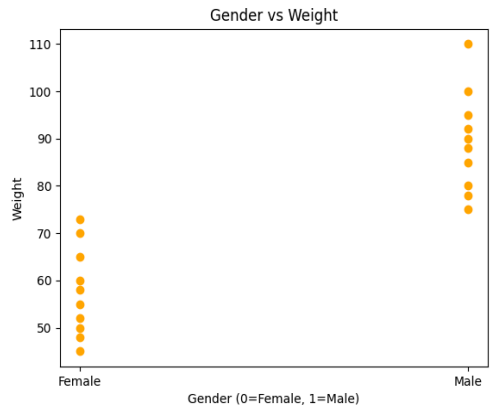
```
=== Matrix Operations ===

Enter rows for Matrix 1: 2
Enter columns for Matrix 1: 2
Enter 4 elements for Matrix 1:
1 2 3 4

Enter rows for Matrix 2: 2
Enter columns for Matrix 2: 2
Enter 4 elements for Matrix 2:
5 6 7 8

Matrix 1:
 [[1 2]
 [3 4]]

Matrix 2:
 [[5 6]
 [7 8]]

Addition Result:
 [[ 6  8]
 [10 12]]

Subtraction Result:
 [[-4 -4]
 [-4 -4]]

Multiplication Result:
 [[19 22]
 [43 50]]
```

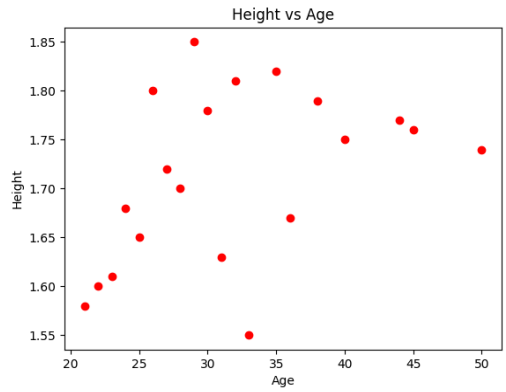
Question 5

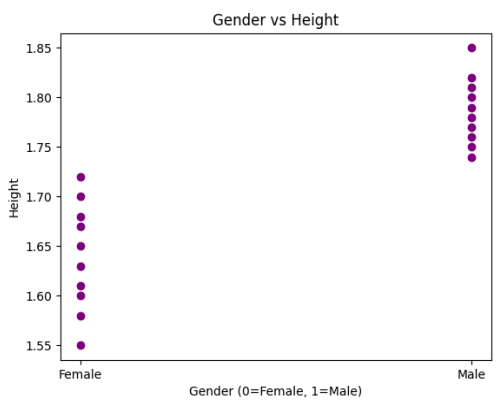
Program Description:

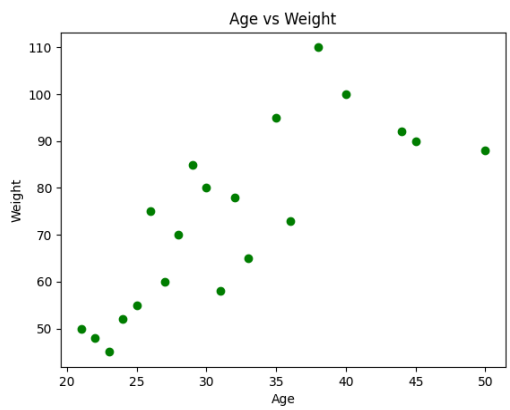
```
This program reads the file health_data.csv using Pandas.It creates
scatter plots to show relationships between variables such as height,
weight, age, and gender using Matplotlib.The plots are displayed and saved
as image files.
```

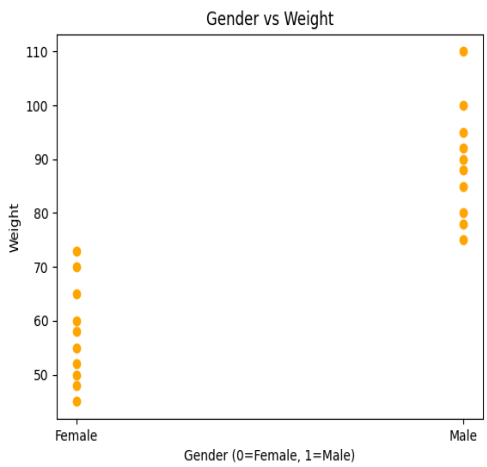
Screenshot:

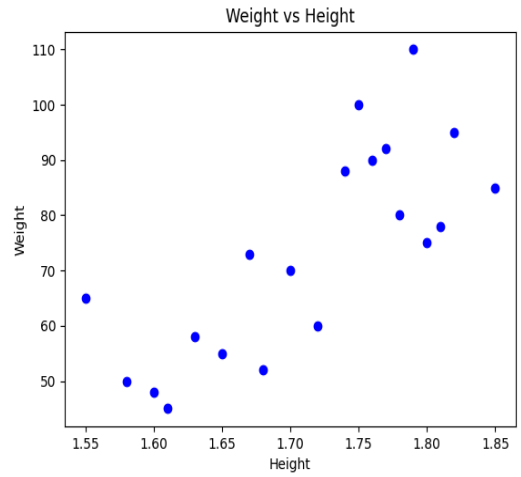













```
Data loaded successfully.
First 5 rows:
     name  age  gender  height  weight
0   Alice   25  Female    1.65      55
1  Jenith   30    Male    1.78      80
2   Carol   22  Female    1.60      48
3   David   35    Male    1.82      95
4    Emma   28  Female    1.70      70

Plot 1 saved: weight vs height
Plot 2 saved: age vs weight
Plot 3 saved: height vs age
Plot 4 saved: gender vs height
Plot 5 saved: gender vs weight

All plots created successfully.

Process finished with exit code 0
```

Question 6

Program Description:

```
This program reads health_data.csv into a Pandas DataFrame and adds two
new columns: BMI (Weight / Height) and Health_Status based on the BMI
value ranges provided.
```

Screenshot:

```
Original data (first 5 rows):
     name  age  gender  height  weight
0   Alice   25  Female    1.65      55
1  Jenith   30    Male    1.78      80
2   Carol   22  Female    1.60      48
3   David   35    Male    1.82      95
4    Emma   28  Female    1.70      70

Updated data with BMI and Health_Status (first 10 rows):
     name  age  gender  ...  weight    BMI                        Health_Status
0   Alice   25  Female  ...      55  33.33  High risk of diabetes/heart disease
1  Jenith   30    Male  ...      80  44.94            Critical health condition
2   Carol   22  Female  ...      48  30.00  High risk of diabetes/heart disease
3   David   35    Male  ...      95  52.20            Critical health condition
4    Emma   28  Female  ...      70  41.18            Critical health condition
5   Frank   40    Male  ...     100  57.14            Critical health condition
6   Grace   31  Female  ...      58  35.58            Critical health condition
7   Henry   26    Male  ...      75  41.67            Critical health condition
8    Isla   24  Female  ...      52  30.95  High risk of diabetes/heart disease
9    Jack   45    Male  ...      90  51.14            Critical health condition

[10 rows x 7 columns]

Updated data saved to health_data_updated.csv

Health Status Summary:
Health_Status
Critical health condition              14
High risk of diabetes/heart disease     5
Overweight risk                         1
Name: count, dtype: int64

Process finished with exit code 0
```